HashTable in C#

Category	Method	Description	Simple Example
➕ Adding	Add(key, value)	Adds a key-value pair	ht.Add("name", "John");
➖ Removing	Remove(key)	Removes item by key	ht.Remove("name");
🧹 Clearing	Clear()	Removes all items	ht.Clear();
🔍 Checking	ContainsKey(key) / Contains(key)	Checks if key exists	ht.ContainsKey("name");
🔍 Checking	ContainsValue(value)	Checks if value exists	ht.ContainsValue("John");
📥 Access	this[key]	Get or set value	ht["name"] = "Jane";
🔁 Iteration	GetEnumerator()	Loop through items	foreach(DictionaryEntry e in ht)
📋 Copying	CopyTo(array, index)	Copy to array	ht.CopyTo(arr, 0);
🧬 Cloning	Clone()	Creates a shallow copy	Hashtable copy = (Hashtable)ht.Clone();
⚖️ Compare	Equals(obj)	Compares two hashtables	ht.Equals(ht2);
🔢 Hash	GetHashCode()	Gets hash code	ht.GetHashCode();
📝 String	ToString()	Converts to string	ht.ToString();

Property	Description	Example
Count	Number of elements	ht.Count
Keys	All keys collection	ht.Keys
Values	All values collection	ht.Values
IsReadOnly	If read-only	ht.IsReadOnly
IsSynchronized	Thread-safe check	ht.IsSynchronized

CRUD → Add, Remove, Clear

Check → ContainsKey, ContainsValue

Access → this[key]

Loop/Copy → GetEnumerator, CopyTo

Other → Clone, Equals, GetHashCode

DICTIONARY IN C#

Category	Method	Description	Simple Example
➕ Adding	Add(key, value)	Adds a key-value pair	dict.Add("name", "John");
➕ Adding	TryAdd(key, value)	Adds if key doesn't exist	dict.TryAdd("age", 20);
➖ Removing	Remove(key)	Removes item by key	dict.Remove("name");
🧹 Clearing	Clear()	Removes all items	dict.Clear();
🔍 Checking	ContainsKey(key)	Checks if key exists	dict.ContainsKey("name");
🔍 Checking	ContainsValue(value)	Checks if value exists	dict.ContainsValue("John");
📥 Access	this[key]	Get or set value	dict["name"] = "Jane";
📥 Safe Access	TryGetValue(key, out value)	Gets value safely (no error if missing)	dict.TryGetValue("name", out var val);
🔁 Iteration	GetEnumerator()	Loop through items	foreach(var kv in dict)
📋 Copying	CopyTo(array, index)	Copy to array	dict.CopyTo(arr, 0);
⚖️ Compare	Equals(obj)	Compares dictionaries	dict.Equals(dict2);
🔢 Hash	GetHashCode()	Gets hash code	dict.GetHashCode();
📝 String	ToString()	Converts to string	dict.ToString();

Property	Description	Example
Count	Number of elements	dict.Count
Keys	Collection of keys	dict.Keys
Values	Collection of values	dict.Values
Item[key]	Access value by key	dict["name"]
Comparer	Key comparer used	dict.Comparer

CRUD → Add, TryAdd, Remove, Clear

Check → ContainsKey, ContainsValue

Access → this[key], TryGetValue

Loop/Copy → GetEnumerator, CopyTo

Other → Equals, GetHashCode

Feature	Dictionary	Hashtable
Namespace	System.Collections.Generic	System.Collections
Type Safety	✅ Strongly typed	❌ Not type-safe
Performance	⚡ Faster	⚡ Fast
Data Type	Specific types (string, int)	object (any type)
Boxing/Unboxing	❌ None	✅ Happens
Usage	Modern C#	Legacy code